

Improved Disturbance Rejection Ability for Speed Control of PMSM Drives Using a Modified Cascaded Extended State Observer

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Abstract—As the demand for high precision in permanent magnet synchronous motors (PMSMs) increases, achieving superior disturbance rejection in PMSM speed regulation becomes essential. In this context, active disturbance rejection control (ADRC) has emerged as a widely employed method. To enhance disturbance rejection, researchers have explored higher order extended state observers (ESOs) and cascaded ESO (CESO). However, the unclear relationship between these approaches leads to redundant efforts, and the fixed ESO structure used in constructing CESO often limits its potential improvement. Additionally, the impact of individual observer bandwidth within CESO on the overall system is poorly understood. To address these gaps, this article analyzes various ESO structures for constructing CESO and identifies two promising CESO configurations with improved disturbance rejection. Subsequently, a comparative analysis of different observer bandwidths is conducted to optimize performance. Furthermore, this article establishes the equivalence between CESO and higher order ESO (HESO). The results are validated through theoretical analysis and experimental tests using dSPACE DS1202 MicroLabBox.

Index Terms—Active disturbance rejection control (ADRC), cascaded extended state observers (ESOs), frequency-domain analysis, permanent magnet synchronous motors (PMSMs).

I. INTRODUCTION

WITH their notable features such as high torque density and high efficiency, permanent magnet synchronous motors (PMSMs) have gained widespread acceptance in high-performance servo applications [1], [2]. However, various

sources of disturbance, including load torque, friction torque, and torque ripple due to cogging torque [3] and inverter nonlinearity [4], [5], can significantly impact the performance of PMSM drives in maintaining reference speed [6]. Consequently, there is a strong demand for designing speed controllers with robust disturbance rejection capabilities.

In this regard, disturbance/uncertainty estimation and attenuation (DUEA) techniques [7], which decouple disturbance rejection from speed tracking, have been developed. Among these techniques, linear active disturbance rejection control (LADRC) which was first proposed by Han [8] and subsequently modified with linear control gains by Gao et al. [9], has garnered significant interest from both academic and industrial communities due to its simplicity as an alternative to classic PI controllers [10] and less dependency on model information [11]. In the LADRC scheme for PMSM speed regulation, proportional feedback control is integrated with an observer which estimates the lumped disturbance term accounting for both external disturbances and internal variations arising from parameter fluctuations or unmodeled dynamics [12]. Consequently, the observer plays a crucial role in the disturbance rejection capability of the entire system.

As a pivotal component of LADRC, significant efforts have been devoted to developing a higher order extended state observer (ESO) to replace the typically used second-order ESO [13] or phase-locking loop observer (PLL) [14]. In [15], a third-order PLL was introduced, resulting in an increased rejection ability for low-frequency disturbances. Subsequently, the observer was extended to the fourth order in [16]. By leveraging the concept of generalized proportional integral observers (GPIOs) [17], both ESOs and PLLs were synthesized as different types of GPIO, enabling a comparison of various observer structures within the same order.

In addition to the advancement of a higher order ESO (HESO), there is a prevailing trend toward adopting a cascaded ESO (CESO). Specifically, in [18], two second-order ESOs were cascaded to enhance system response under disturbance. The first ESO estimates a portion of the disturbance, leading to a reduced response time for the subsequent ESO. To further improve disturbance rejection ability against periodic disturbances, Cao et al. [19] introduced a quasi-general integrator into the CESO. Additionally, in [20] and [21], CESOs were

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linked in parallel to achieve better noise suppression. However, this configuration is often considered overly complex.

Findings from the existing literature consistently demonstrate a common trend: compared to the conventional ESO, both integrating an ESO that is one order higher and linking an additional ESO in cascade can enhance disturbance rejection or noise suppression [22]. Furthermore, certain studies suggest a potential equivalence between HESOs and CESOs. For instance, in [23], a fourth-order HESO was transformed into a three-layered CESO using second-order ESOs. Subsequently in [24], the three-layered CESO was streamlined to a two-layered configuration indicating the equivalence among them.

However, a clear relationship between HESOs and CESOs has not yet been established. This lack of clarity can lead to redundant efforts, as newly proposed CESO structures may essentially replicate the functionalities of previously developed HESOs. Additionally, the fixed ESO structure employed in CESO methodologies often limits advancements in disturbance rejection. Moreover, the influence of individual observer bandwidth within the CESO on the overall system remains ambiguous. To address these gaps, this article analyzes various types of ESOs for constructing CESOs and identifies two promising CESO configurations that demonstrate enhanced disturbance rejection compared to conventional setups. Furthermore, a comparative analysis is conducted to assess the impact of different observer bandwidths on achieving superior disturbance rejection capability.

In conclusion, the contributions of this article can be summarized as follows.

- 1) Two CESO configurations are proposed with enhanced disturbance rejection compared to conventional setups.
- 2) The optimal selection of observer bandwidth in CESO for superior disturbance rejection capabilities is identified.
- 3) The equivalence of CESOs and HESOs is revealed and demonstrated using the example of a conventional two-layered CESO.

The remainder of the article is structured as follows. Section II details the procedure of designing LADRC for PMSM speed regulation using the conventional CESO and identifies its equivalent HESO. Subsequently in Section III, modified CESO configurations for enhanced disturbance rejection ability are proposed, along with analyses on optimal observer bandwidth selection. Section IV presents the experimental validation of the findings. Finally, conclusions are drawn in Section V.

II. LADRC FOR PMSM SPEED REGULATION USING THE CONVENTIONAL CESO

A. PMSM Mechanical Model

The mechanical model of PMSM is governed by Newton's second law of motion which is

$$\dot{\Omega} = \frac{-B\Omega + T_e - T_L}{J} = bT_e + d_{ext} \quad (1)$$

where Ω is the rotor mechanical angular speed, B is the viscous friction torque coefficient, T_e is the electrical torque,

T_L is the load torque, J is the rotor inertia, coefficient b is defined as $b = 1/J$, and external disturbance $d_{ext} = (-B\Omega - T_L)/J$ is caused by the load torque and viscous friction torque.

However, with the operation of PMSM, its rotor inertia J is not always equal to its nominal value J_n and some error may appear between the commanded electrical torque T_e^* and its actual value T_e . Hence, by denoting $b_n = 1/J_n$, the mechanical model (1) can be modified as

$$\dot{\Omega} = b_n T_e^* + d_{to} \quad (2)$$

where $d_{to} = d_{ext} + b(T_e - T_e^*) + (b - b_n)T_e^*$ is the total disturbance including both the internal disturbances caused by the torque tracking error and inertia mismatches and the external disturbance d_{ext} .

B. LADRC Using the Conventional ESO

The tracking error e_s between the commanded speed Ω^* and its actual value Ω can be expressed as

$$e_s = \Omega^* - \Omega. \quad (3)$$

And according to (2), the state equation of the speed tracking error e_s can be obtained as

$$\dot{e}_s = \dot{\Omega}^* - (b_n T_e^* + d_{to}). \quad (4)$$

Then, designing the tracking error convergence law in a simple proportional control with control gain k_p as

$$\dot{e}_s = -k_p e_s. \quad (5)$$

Subsequently substituting (5) into (4) yields the output of the controller as

$$T_e^* = \frac{\dot{\Omega}^* + k_p (\Omega^* - \Omega) - d_{to}}{b_n}. \quad (6)$$

In addition, between the commanded torque and its actual value, there exists a torque control loop which works as a first-order inertia process with a time constant of τ as

$$D(s) = \frac{1}{\tau s + 1}. \quad (7)$$

Also, consider the speed measurement noise δ_s which functions as

$$\Omega^m = \Omega + \delta_s \quad (8)$$

where Ω^m is the measured speed.

Based on (2), a classical second-order ESO can be built as (9) to estimate the speed and disturbance

$$\begin{cases} \tilde{\Omega} = \Omega^m - \hat{\Omega} \\ \dot{\hat{\Omega}} = b_n T_e^* + \hat{d}_{to} + k_1 \tilde{\Omega} \\ \dot{\hat{d}}_{to} = k_2 \tilde{\Omega} \end{cases} \quad (9)$$

where $\hat{\Omega}$ and $\hat{d}_{to}$ are the estimated values of Ω^m and d_{to} , k_1 and k_2 are the observer gains.

Using the estimated states, the torque reference in (6) can be modified as

$$T_e^* = \frac{s\Omega^* + k_p (\Omega^* - \hat{\Omega}) - \hat{d}_{to}}{b_n}. \quad (10)$$

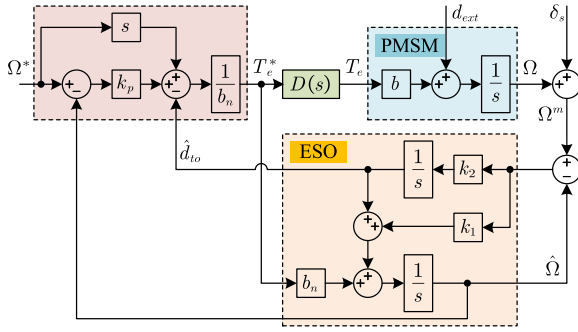


Fig. 1. Block diagram of LADRC for PMSM speed regulation.

Finally, the entire control system can be built as shown in Fig. 1.

C. LADRC Using the Conventional CESO

To avoid misunderstanding and separate those two observers in CESO, the two control gains k_1 and k_2 for the observers in each layer are renamed as k_{i1} and k_{i2} , where i stands for the number of layers in CESO.

Also, with the concept of GPIO, the estimated speed and disturbance of the observer can be acquired from different branches. Hence, G_{i1} and G_{i2} are used to represent the transfer functions of different selections of observer in the i th layer. G_{i1} can be selected as either 0 or 1 while G_{i2} can be chosen between k_{i2}/s and $k_{i1} + k_{i2}/s$.

The ESO in the first layer of CESO is built as

$$\begin{cases} \tilde{\Omega}_1 = \Omega^m - \hat{\Omega}_1 \\ \dot{\tilde{\Omega}}_1 = b_n T_e^* + \left(k_{11} + \frac{k_{12}}{s}\right) \tilde{\Omega}_1 \\ \Omega_1 = \Omega^m - G_{11} \tilde{\Omega}_1 \\ \hat{d}_{to1} = G_{12} \tilde{\Omega}_1 \end{cases} \quad (11)$$

where $\hat{\Omega}_1$ and $\hat{d}_{to1}$ are the estimated speed and estimated total disturbance by the first ESO while Ω_1 is its output speed.

Subsequently, the ESO in the second layer built as

$$\begin{cases} \tilde{\Omega}_2 = \Omega_1 - \hat{\Omega}_2 \\ \dot{\tilde{\Omega}}_2 = b_n T_e^* + \left(k_{21} + \frac{k_{22}}{s}\right) \tilde{\Omega}_2 + \hat{d}_{to1} \\ \Omega_2 = \Omega_1 - G_{21} \tilde{\Omega}_2 \\ \hat{d}_{to2} = G_{22} \tilde{\Omega}_2 \end{cases} \quad (12)$$

where $\hat{\Omega}_2$ and $\hat{d}_{to2}$ are the estimated speed and estimated total disturbance by the second ESO while Ω_2 is its output speed.

The final output disturbance of CESO is the summation of the estimated disturbances from two layers while the output speed is the speed from the second layer. Although the output speed may be the measured value in some cases, to unify its expression, it is denoted as $\hat{\Omega}$. Hence, the output speed $\hat{\Omega}$ and total disturbance $\hat{d}_{to}$ of CESO can be expressed as

$$\begin{cases} \hat{\Omega} = \Omega_2 \\ \hat{d}_{to} = \hat{d}_{to1} + \hat{d}_{to2}. \end{cases} \quad (13)$$

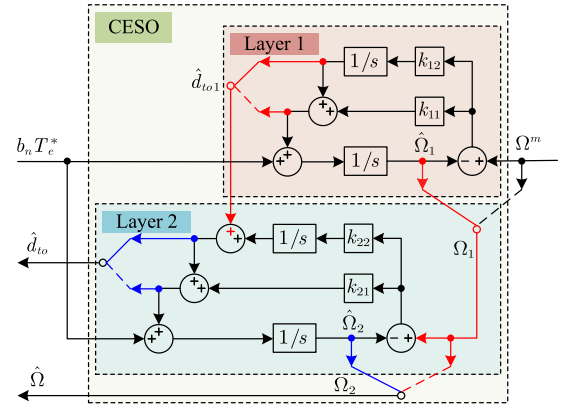


Fig. 2. Block diagram of CESO using two second-order ESOs.

Finally, the CESO is built as Fig. 2 and by using the selection shown in Fig. 2, it represents the conventional CESO.

By sequentially substituting (11) into (12) and (13), the final output of CESO can be obtained as

$$\begin{cases} \hat{\Omega} = \frac{\sigma_1 \Omega^m + \sigma_2 s b_n T_e^*}{\Delta_{o1} \Delta_{o2}} \\ \hat{d}_{to} = \frac{\sigma_3 s^2 \Omega^m - \sigma_3 s b_n T_e^*}{\Delta_{o1} \Delta_{o2}} \end{cases} \quad (14)$$

where σ_1 , σ_2 , and σ_3 as listed in the Appendix are polynomials of G_{ij} and k_{ij} . Δ_{oi} as shown in (15) are the characteristic polynomial of each observer

$$\Delta_{oi} = s^2 + k_{i1}s + k_{i2}. \quad (15)$$

To place the poles, the control gains are usually set as

$$\begin{cases} k_{i1} = 2\omega_{oi} \\ k_{i2} = \omega_{oi}^2 \end{cases} \quad (16)$$

where ω_{oi} is the control bandwidth of the observer in the i th layer of CESO.

From (14), it can be seen that the characteristic equation of CESO is a fourth-order polynomial. And because as a fourth-order ESO, the transfer function of its error introduced in [16] is

$$\tilde{\Omega} = \frac{s^4 \Omega^m - s^3 b_n T_e^*}{s^4 + k_1 s^3 + k_2 s^2 + k_3 s + k_4} \quad (17)$$

where the observer gains in the denominator are usually set with the same principle as (16) which is: $k_1 = 4\omega_o$, $k_2 = 6\omega_o^2$, $k_3 = 4\omega_o^3$, and $k_4 = \omega_o^4$.

And by substituting (17) into (14), it can be found that in the situation of

$$\begin{cases} k_1 = k_{11} + k_{21} \\ k_2 = k_{12} + k_{22} + k_{11}k_{21} \\ k_3 = k_{11}k_{22} + k_{12}k_{21} \\ k_4 = k_{12}k_{22} \end{cases} \quad (18)$$

the conventional CESO is equivalent to a fourth-order ESO with output as (19). By using the pole placement method to determine the observer gains as introduced above, it is evident

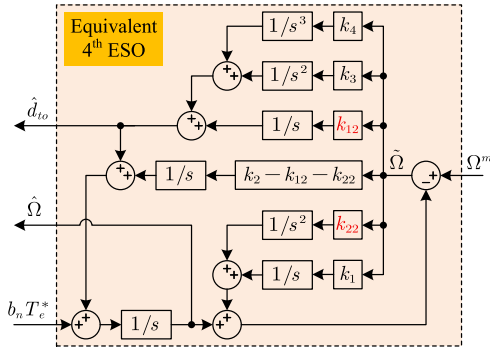


Fig. 3. Block diagram of the equivalent fourth-order ESO.

that (18) always holds. Based on (17) and (19), the equivalent fourth-order ESO is built in Fig. 3

$$\begin{cases} \hat{\Omega} = \Omega^m - \left(1 + \frac{k_1}{s} + \frac{k_{22}}{s^2}\right) \tilde{\Omega} \\ \hat{d}_{io} = \left(\frac{k_{12}}{s} + \frac{k_3}{s^2} + \frac{k_4}{s^3}\right) \tilde{\Omega}. \end{cases} \quad (19)$$

To be noted, (18) reveals that k_1 , k_2 , k_3 , and k_4 are uniformly influenced by both observers from the two layers in the CESO. However, as depicted in (19), k_{12} is distinctly involved in disturbance estimation, while k_{22} plays a separate role in speed estimation. This observation suggests that the first observer primarily dictates the disturbance estimation ability, whereas the second observer predominantly governs the speed estimation property.

III. PROPOSED CESOS FOR DISTURBANCE REJECTION ABILITY IMPROVEMENT

A. Analysis on Second-Order ESOs

Since the core of the observers is to estimate the disturbance and speed properly. Two transfer functions including $\phi_d = \hat{d}_{io}/d_{io}$ and $\phi_\Omega = \hat{\Omega}/\Omega^m$ representing each property are defined. Between them, ϕ_d can be obtained by substituting (11) and (2) into its expression which results in

$$\phi_d = \frac{sG_{12}}{\Delta_{o1}}. \quad (20)$$

While ϕ_Ω as shown in (21) can be intuitively obtained by (11) without $b_n T_e^*$

$$\phi_\Omega = 1 - \frac{s^2 G_{11}}{\Delta_{o1}}. \quad (21)$$

Since both the estimated speed and disturbance offer two options, there exist four variations of second-order ESOs, denoted as 2nd ESO_k, where $k = 1, 2, 3, 4$. These configurations can be constructed by substituting the respective G_{ij} values from Table I into (11) and (12). Among these four kinds of ESO, 2nd ESO₁ corresponds to the classical second-order ESO presented in (9) and Fig. 1.

Using (20) and (21), the bode plots of disturbance and speed estimation for second-order ESOs can be obtained as Fig. 4 with the parameters setting as $\tau = 2 \times 10^{-4}$ s, $k_p = 20\pi$ rad/s and $\omega_{o1} = 200\pi$ rad/s. k_{11} and k_{12} are determined by (16).

TABLE I
TRANSFER FUNCTIONS G_{ij} OF SECOND-ORDER ESOs

Type	G_{11}	G_{12}	G_{21}	G_{22}
2 nd ESO ₁	1	k_{12}/s	0	0
2 nd ESO ₂	1	$k_{11} + k_{12}/s$	0	0
2 nd ESO ₃	0	k_{12}/s	0	0
2 nd ESO ₄	0	$k_{11} + k_{12}/s$	0	0

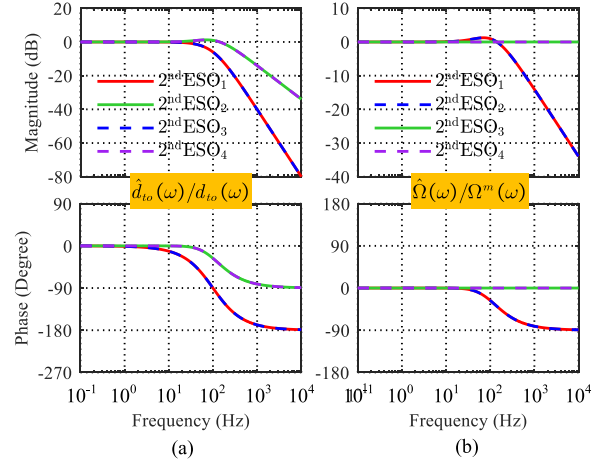


Fig. 4. Bode plots of estimation abilities of different second-order ESOs. (a) Disturbance estimation. (b) Speed estimation.

Fig. 4(a) illustrates that all four ESO configurations are capable of adequately estimating disturbances in the low-frequency range. However, as the disturbance frequency increases, the estimation ability of 2nd ESO₁ and 2nd ESO₃ rapidly declines at a rate of -40 dB/dec. In contrast, 2nd ESO₂ and 2nd ESO₄ exhibit a slower decline at a rate of -20 dB/dec, indicating their superior disturbance estimation capability.

Since 2nd ESO₃ and 2nd ESO₄ employ the measured speed as their estimated speed, it results in a zero slope straight line in Fig. 4(b). Consequently, 2nd ESO₃ and 2nd ESO₄ lack suppression property for measurement noise. Conversely, 2nd ESO₁ and 2nd ESO₂ demonstrate the capability to mitigate high-frequency noise at a rate of -20 dB/dec.

Subsequently, to examine the effect of the observers on the whole system, three transfer functions including $\Phi_r = \Omega/\Omega^*$, $\Phi_d = \Omega/d_{ext}$, and $\Phi_n = \Omega/\delta_s$ are defined to represent tracking performance, disturbance rejection ability, and noise suppression property, respectively. They can be obtained by the following procedures.

Substituting (14) into (10) yields

$$b_n T_e^* = \frac{\sigma_4 \Omega^* - \sigma_5 \Omega^m}{\sigma_6} \quad (22)$$

where σ_4 , σ_5 and σ_6 are the simplified coefficients listed in the Appendix.

Then, considering the effect of the torque control loop in (7) yields

$$b_n T_e^* = \frac{s\Omega^m - s\delta_s - d_{ext}}{D(s)}. \quad (23)$$

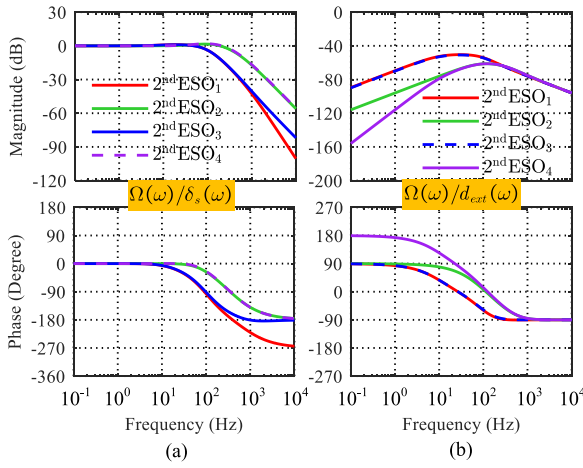


Fig. 5. Bode plots of different second-order ESOs. (a) Noise suppression property. (b) Disturbance rejection ability.

And substituting (23) into (22) yields

$$\Delta_{cl}\Omega^m = D(s)\sigma_4\Omega^* + \sigma_6s\delta_s + \sigma_6d_{ext} \quad (24)$$

where Δ_{cl} is the characteristic equation of the system shown as

$$\Delta_{cl} = \sigma_6s + D(s)\sigma_5. \quad (25)$$

To extract actual speed Ω from its measured value Ω^m , (8) is substituted into (24) which yields

$$\Omega = \frac{D(s)\sigma_4}{\Delta_{cl}}\Omega^* + \frac{\sigma_6}{\Delta_{cl}}d_{ext} + \frac{\sigma_6s - \Delta_{cl}}{\Delta_{cl}}\delta_s. \quad (26)$$

Finally, the transfer functions of speed response relating to speed reference, external disturbance, and speed measurement noise are obtained as

$$\begin{cases} \Phi_r = \frac{D(s)\sigma_4}{\Delta_{cl}} \\ \Phi_d = \frac{\sigma_6}{\Delta_{cl}} \\ \Phi_n = \frac{\sigma_6s}{\Delta_{cl}} - 1 = s\Phi_d - 1. \end{cases} \quad (27)$$

From (27), it shows a coupling between noise suppression and disturbance rejection, and based on (27), the bode plots representing those two properties can be drawn as Fig. 5.

From Fig. 5, it can be seen that, compared to the conventional used second-order ESO which is 2nd ESO₁, 2nd ESO₄ increases the slope of disturbance rejection from 20 to 40 dB/dec indicating a less speed variation encountering external disturbance. Hence, to improve the disturbance rejection ability of the conventional CESO, 2nd ESO₄ is proposed to replace 2nd ESO₁ in the conventional CESO configuration.

In conclusion, the properties of different second-order ESOs shown in Figs. 4 and 5 are summarized in Table II where S_{ϕ_d} , S_{Φ_d} , S_{ϕ_Ω} , and S_{Φ_n} are donated as the slopes of each bode curves.

TABLE II
DYNAMIC PERFORMANCE OF SYSTEM WITH SECOND-ORDER ESOs

ESO	Disturbance Related		Noise Related	
Type	S_{ϕ_d} (dB/dec)	S_{Φ_d} (dB/dec)	S_{ϕ_Ω} (dB/dec)	S_{Φ_n} (dB/dec)
2 nd ESO ₁	-40	20	-20	-60
2 nd ESO ₂	-20	20	-20	-40
2 nd ESO ₃	-40	20	0	-40
2 nd ESO ₄	-20	40	0	-40

Note: S_{ϕ_d} , S_{Φ_d} , S_{ϕ_Ω} and S_{Φ_n} represent the slopes of bode curves indicating disturbance estimation, disturbance rejection, speed estimation and noise suppression, respectively.

TABLE III
TRANSFER FUNCTIONS G_{ij} OF CESOS

Type	G_{11}	G_{12}	G_{21}	G_{22}
CESO ₁₁	1	k_{12}/s	1	k_{12}/s
CESO ₄₄	0	$k_{11} + k_{12}/s$	0	$k_{11} + k_{12}/s$
CESO ₁₄	1	k_{12}/s	0	$k_{11} + k_{12}/s$
CESO ₄₁	0	$k_{11} + k_{12}/s$	1	k_{12}/s

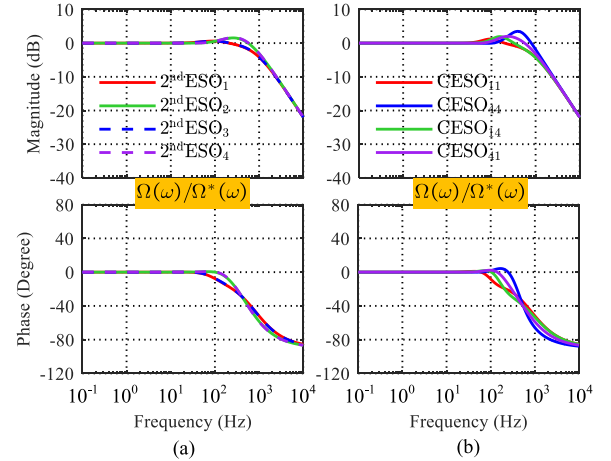


Fig. 6. Tracking performance of different observers. (a) Second-order ESOs. (b) CESOs.

B. Analysis on CESOs

To verify the disturbance rejection enhancement of replacing 2nd ESO₁ with 2nd ESO₄, different CESOs are compared which are denoted as CESO_{xy} where x stands for the type of ESO in the first layer while y stands for the type of ESO in the second layer. Their transfer functions are listed in Table III. Among them, CESO₁₁ is the conventional CESO. CESO₄₄, CESO₁₄, and CESO₄₁ are the modified CESOs with different estimated disturbance and speed selections in Fig. 2. The parameters are set as: $\tau = 2 \times 10^{-4}$ s, $k_p = 20\pi$ rad/s, $\omega_{o1} = \omega_{o2} = 200\pi$ rad/s.

Based on (27), the bode plots representing tracking performance, disturbance rejection ability, and noise suppression property of different CESOs can be drawn in Figs. 6 and 7.

Fig. 6 compares the tracking performance of CESOs and second-order ESOs, and it verifies that by using CESOs,

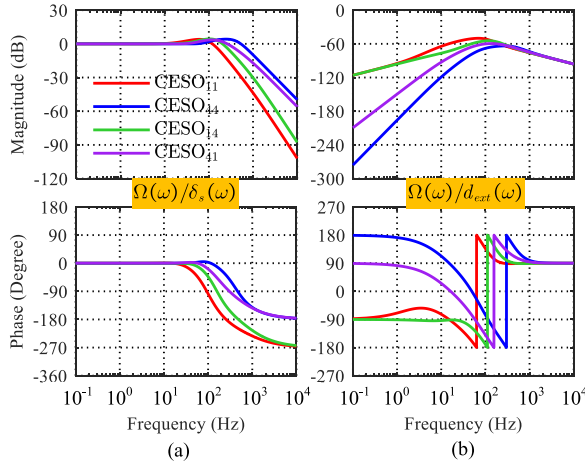


Fig. 7. Bode plots of different CESOs. (a) Noise suppression property. (b) Disturbance rejection ability.

TABLE IV
DYNAMIC PERFORMANCE OF SYSTEM WITH CESOS

CESO Type	Noise Suppression S_{Φ_n} (dB/dec)	Disturbance Rejection S_{Φ_d} (dB/dec)
CESO ₁₁	-60	20
CESO ₄₄	-40	80
CESO ₁₄	-60	20
CESO ₄₁	-40	60

the tracking performance will not deteriorate. However, the bode plot of CESO₄₄ reveals a notable amplification effect in the medium frequency range, potentially leading to a larger overshoot in speed response. Moreover, given its direct utilization of measured speed, the observer exhibits poor noise suppression property. Consequently, the usage of CESO₄₄ is not recommended within the context of the article.

Fig. 7 shows that by using 2nd ESO₄ with 2nd ESO₁, the disturbance rejection ability is improved. Specifically, the proposed CESO₁₄ have better disturbance rejection ability than the conventional CESO₁₁ in the frequency range from 1 to 100 Hz and the proposed CESO₄₁ improve the rejection rate from 20 to 60 dB/dec which enables it to eliminate the effect of a ramp disturbance.

In conclusion, the disturbance rejection ability and noise suppression property of different CESOs shown in Fig. 7 are summarized in Table IV.

C. Analysis on Different Control Gain Settings

To verify how the two controller bandwidths will affect the system, CESO₁₁ are examined with different control gains which are set as the following four cases: 1) the original setup: $\omega_{o1} = \omega_{o2} = \omega_o$; 2) increasing both of the observer bandwidths equally: $\omega_{o1} = \omega_{o2} = 2\omega_o$; 3) only increasing the former observer bandwidth: $\omega_{o1} = 4\omega_o$, $\omega_{o2} = \omega_o$; and 4) only increasing the latter observer bandwidth: $\omega_{o1} = \omega_o$, $\omega_{o2} = 4\omega_o$. The bode plots of these four cases are drawn in Fig. 8.

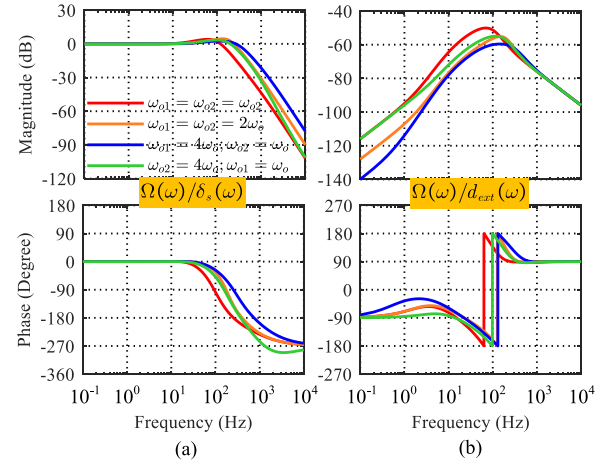


Fig. 8. Bode plots of CESO₁₁ with different observer bandwidths. (a) Noise suppression property. (b) Disturbance rejection ability.

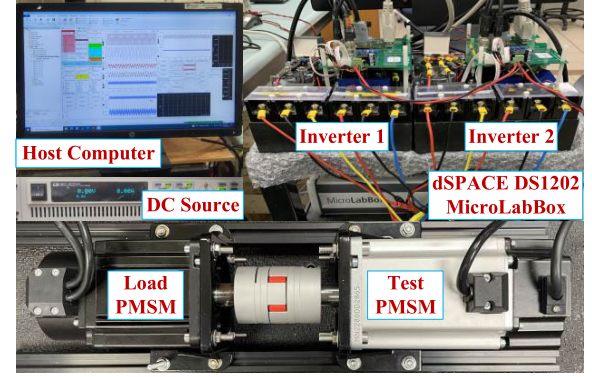


Fig. 9. Test bench.

TABLE V
PARAMETERS OF THE TEST PMSM

Symbol	Quantity	Symbol	Quantity
Rated power	1 kW	Rated speed	3000 r/min
Rated voltage	220 V	Rated torque	3.2 Nm
d axis inductance	2.66 mH	q axis inductance	4.44 mH
Number of poles	10	Stator resistance	0.59 Ohm
Rotor flux	0.06 Wb	Motor inertia	5.58×10^{-4} kgm ²

Fig. 8 shows that increasing the bandwidth of the first observer appears to have the most disturbance rejection improvement followed by an equal increase in both bandwidths and only increases the latter ESO. It supports the finding that the first observer primarily dictates the disturbance estimation ability.

IV. EXPERIMENTAL RESULTS

A. Test Bench Setup

To verify the findings, a test bench is set up as shown in Fig. 9. The dc bus voltage is set as 180 V with a maximum dc current of five amperes. The parameters of the test PMSM are listed in Table V.

Then, three groups of tests are conducted in the following subsectors aiming at: 1) examining the disturbance rejection

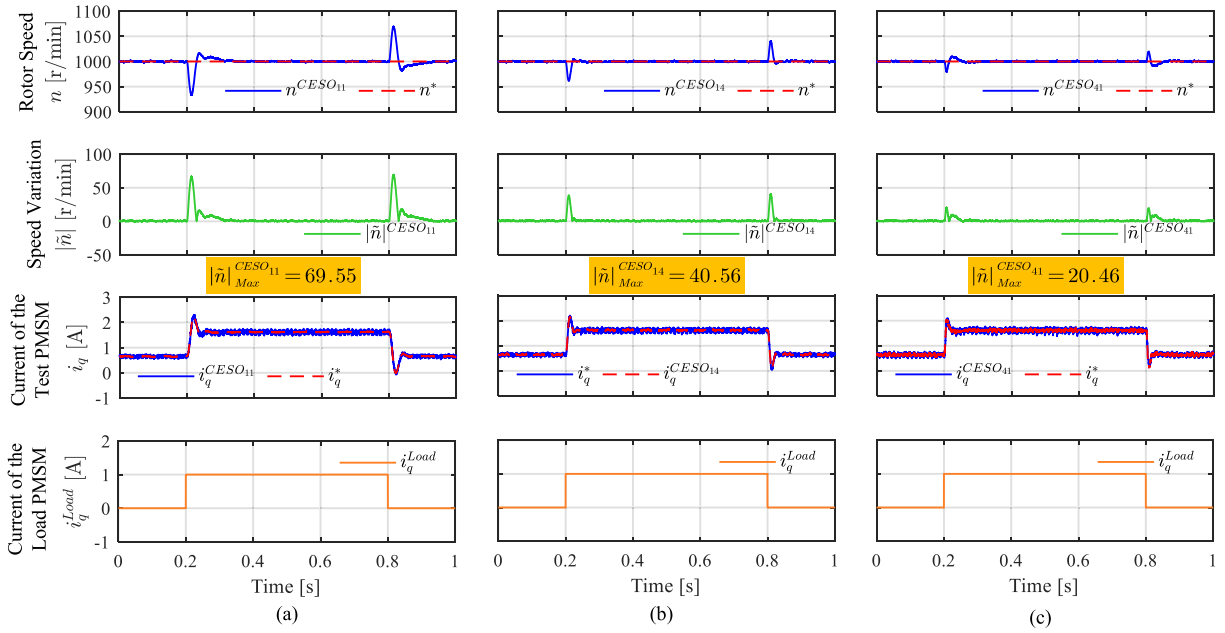


Fig. 10. System response under 1000 r/min with a step load change using different CESOs. (a) Conventional CESO₁₁. (b) Proposed CESO₁₄. (c) Proposed CESO₄₁.

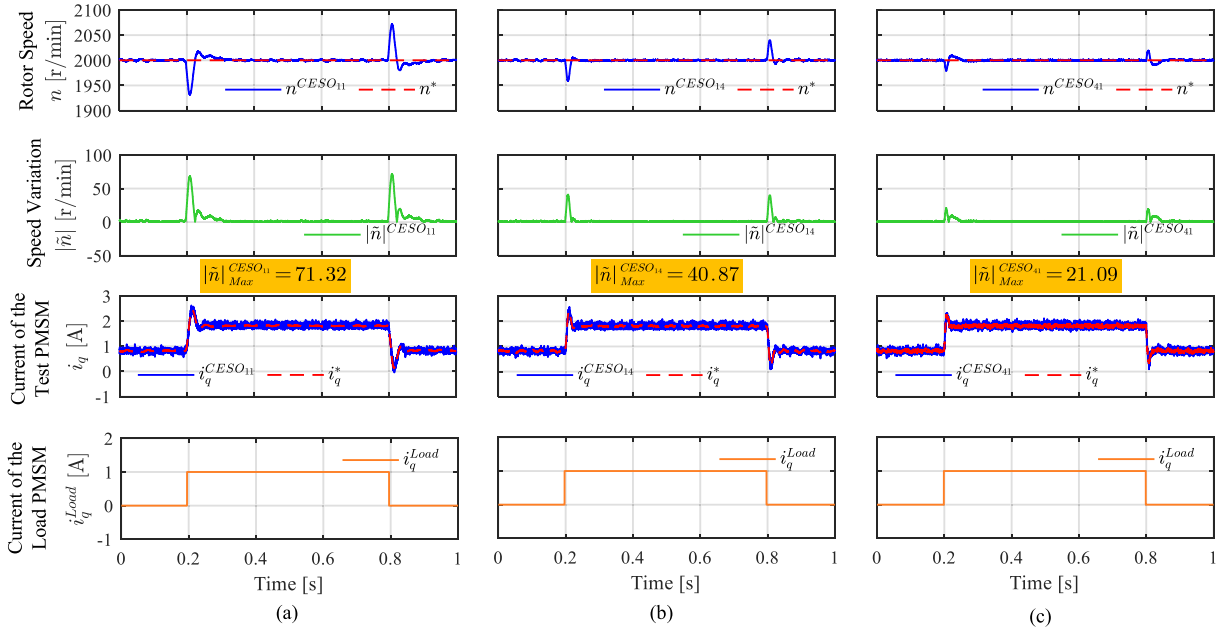


Fig. 11. System response under 2000 r/min with a step load change using different CESOs. (a) Conventional CESO₁₁. (b) Proposed CESO₁₄. (c) Proposed CESO₄₁.

enhancement of the proposed CESO₁₄ and CESO₄₁ compared to the conventional CESO₁₁; 2) revealing the bandwidth of the former ESO contributes more to the disturbance rejection improvement; and 3) proving the equivalence of CESO and HESO using the example of conventional CESO₁₁ and fourth-order ESO shown in Fig. 3.

The control gains in experiments are set as: $k_p = 10\pi$ rad/s and $\omega_o = 20\pi$ rad/s. Speed response, absolute speed variation, and current response of the test PMSM, together with the current given to the load PMSM are recorded in the following figures.

B. Identification of CESO Configurations With Enhanced Disturbance Rejection Compared to Conventional Setups

To examine the enhancement of disturbance rejection ability, a step current of 1 A was given to the load PMSM to generate a step load change at 0.2 s and unloaded at 0.8 s. Figs. 10–12 record the system response of the test PMSM rotating at 1000, 2000, and 3500 r/min, respectively, with different CESOs.

In addition to the above medium-high speed test, Fig. 13 records the system response using CESO₁₁, CESO₁₄ and CESO₄₁ under 500 r/min. A trapezoidal current with an absolute ramp rate of 50 A/s and a maximum value of 5 A

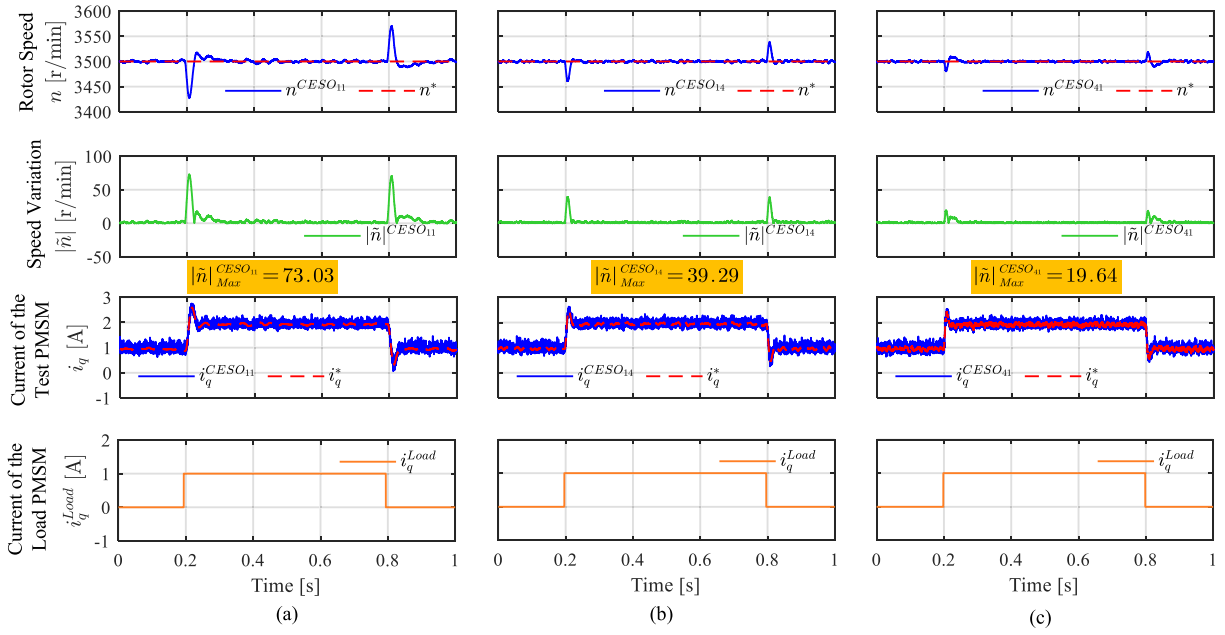


Fig. 12. System response under 3500 r/min with a step load change using different CESOs. (a) Conventional CESO₁₁. (b) Proposed CESO₁₄. (c) Proposed CESO₄₁.

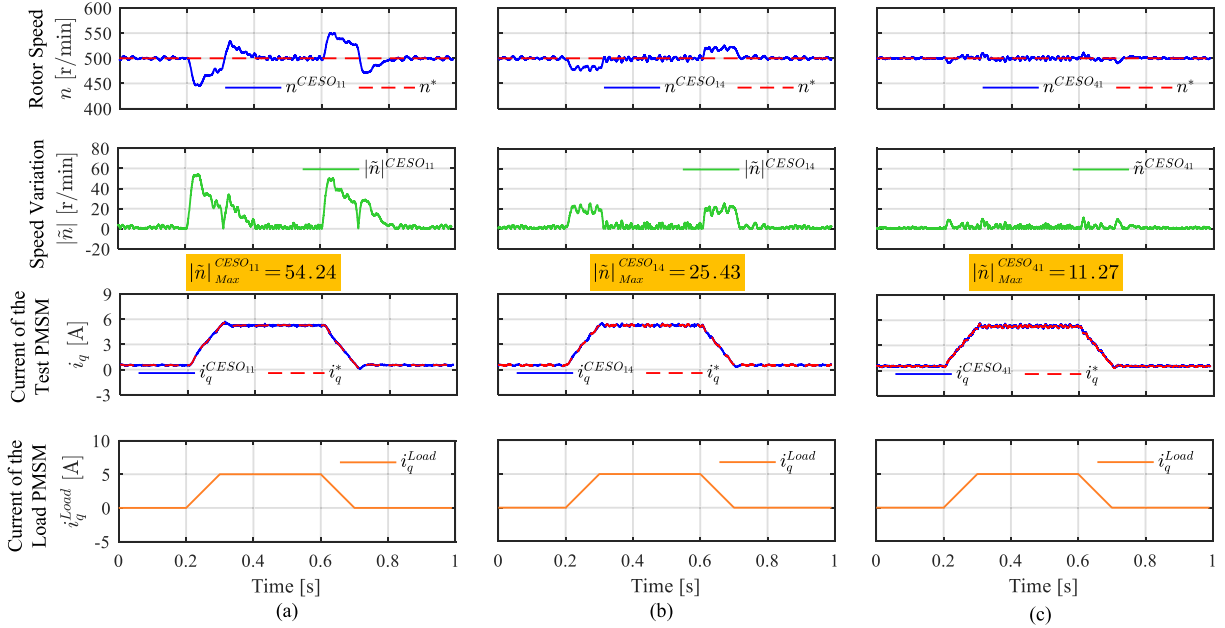


Fig. 13. System response under 500 r/min with a ramp load change using different CESOs. (a) Conventional CESO₁₁. (b) Proposed CESO₁₄. (c) Proposed CESO₄₁.

was given to the load PMSM to mimic a ramp load change starting at 0.2 s and decreasing at 0.6 s.

Figs. 10–12 reveal a consistent trend: the proposed CESO₁₄ reduces speed variation due to a step load change by approximately 30 r/min compared to the conventional CESO₁₁, while the proposed CESO₄₁ achieves an additional reduction of around 20 r/min. These results demonstrate that CESO₁₄ and CESO₄₁ significantly enhance disturbance rejection in response to step load changes within the medium-to-high speed range.

Furthermore, Fig. 13 corroborates their effectiveness against ramp load changes in the low-speed range. And their reduction

compared to the conventional setup is 28.81 and 42.97 r/min, respectively. Therefore, it can be concluded that the proposed methods are effective across the whole speed range.

C. Selection of Observer Bandwidth for Superior Disturbance Rejection Capabilities

To examine the dominant effect of the first observer bandwidth on disturbance rejection enhancement, CESO₁₁ was tested with varying observer bandwidth selections under 1000 r/min with a step load change and 500 r/min with a ramp load change. The system responses of those two tests are recorded in Figs. 14 and 15.

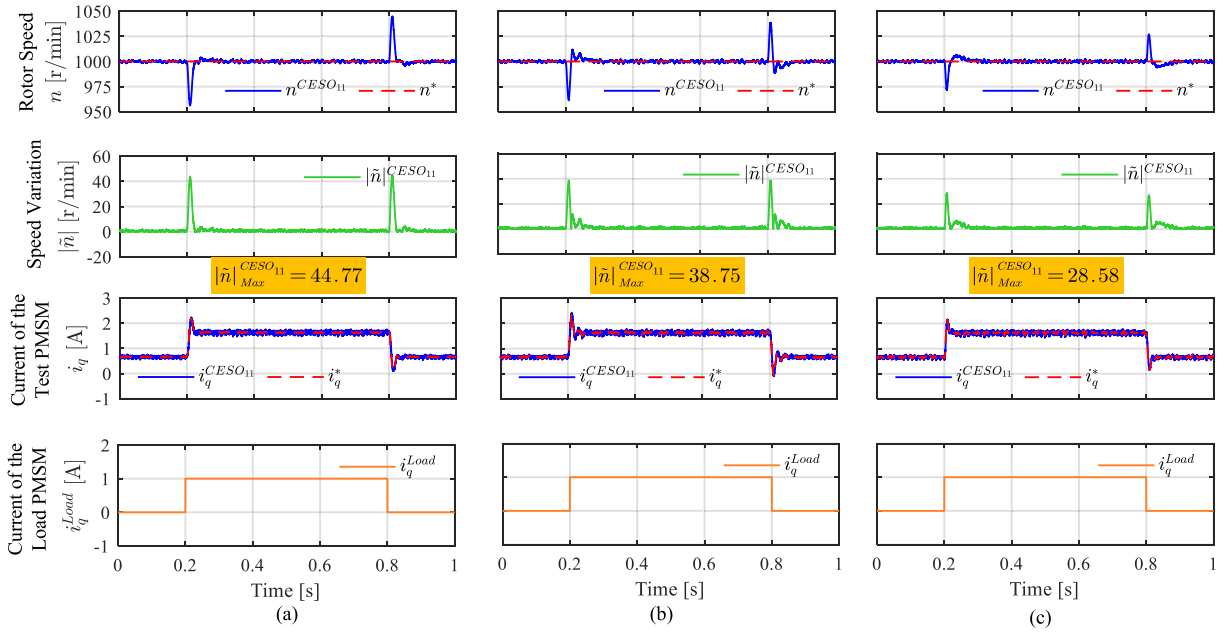


Fig. 14. System response using CESO₁₁ under 1000 r/min with a step load change. (a) $\omega_{o1} = \omega_o$, $\omega_{o2} = 4\omega_o$. (b) $\omega_{o1} = \omega_{o2} = 2\omega_o$. (c) $\omega_{o1} = 4\omega_o$, $\omega_{o2} = \omega_o$.

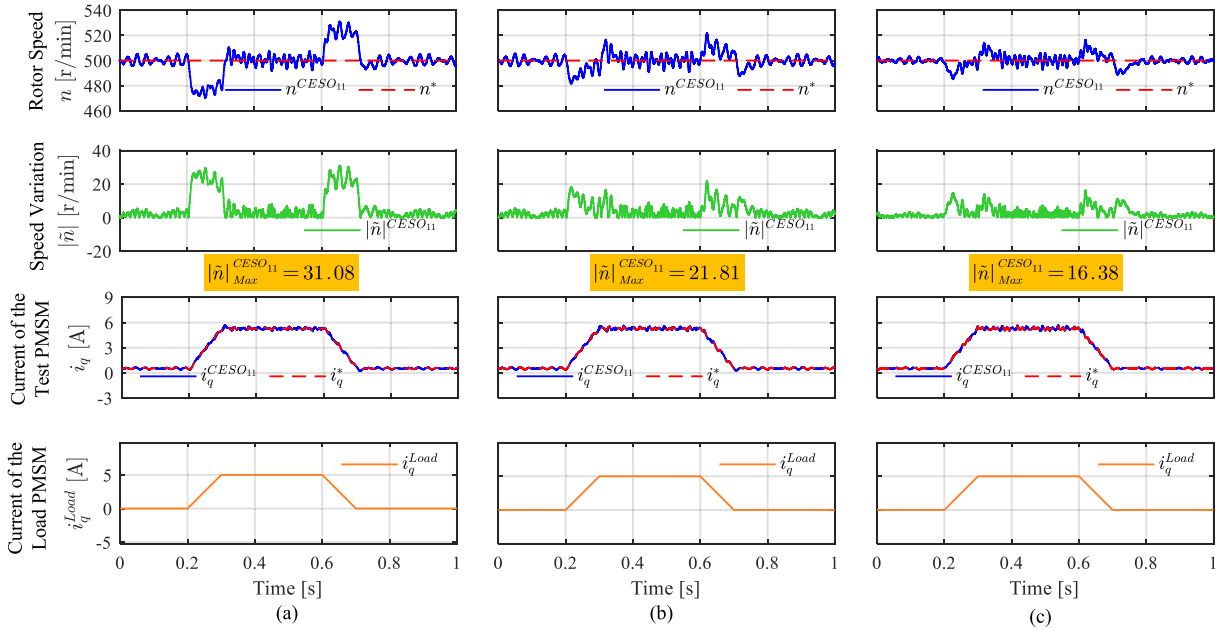


Fig. 15. System response using CESO₁₁ under 500 r/min with a ramp load change. (a) $\omega_{o1} = \omega_o$, $\omega_{o2} = 4\omega_o$. (b) $\omega_{o1} = \omega_{o2} = 2\omega_o$. (c) $\omega_{o1} = 4\omega_o$, $\omega_{o2} = \omega_o$.

From Fig. 14, it can be seen that increasing the first observer bandwidth by a factor of four results in a speed variation of 28.58 r/min, which is smaller than the variation observed when doubling both observer bandwidths (38.75 r/min) or increasing the second observer bandwidth by a factor of four (44.77 r/min). This demonstrates that increasing the first observer bandwidth more effectively improves the disturbance rejection ability of CESO. This conclusion is further supported by Fig. 15 where the speed variations are 16.38, 21.81, and 31.08 r/min, respectively.

Finally, all the speed variations of the experiments under 1000 r/min are listed in Table VI. From Table VI, it can be

seen that the proposed CESO₄₁ can improve the disturbance rejection ability even compared with the conventional CESO having twice its observer bandwidth. Moreover, increasing the bandwidth of the former observer in CESO is more effective in achieving better disturbance rejection ability.

D. Equivalence of CESO and High-Order ESO

To verify the equivalence between CESO and HESO. The example of CESO₁₁ and its equivalent fourth-order ESO as shown in Fig. 3 are tested under 1000 r/min with a step load change. System responses are recorded in Fig. 16.

TABLE VI
SUMMARY OF THE SPEED VARIATIONS IN THE
EXPERIMENTS UNDER 1000 r/min

Type	ω_{o1} (rad/s)	ω_{o2} (rad/s)	Step load (r/min)	Ramp load (r/min)
CESO ₄₁	20 π	20 π	20.46	11.27
CESO ₁₄	20 π	20 π	40.56	25.43
CESO ₁₁	20 π	20 π	69.55	54.24
CESO ₁₁	20 π	80 π	44.77	31.08
CESO ₁₁	40 π	40 π	38.75	21.81
CESO ₁₁	80 π	20 π	28.58	16.38

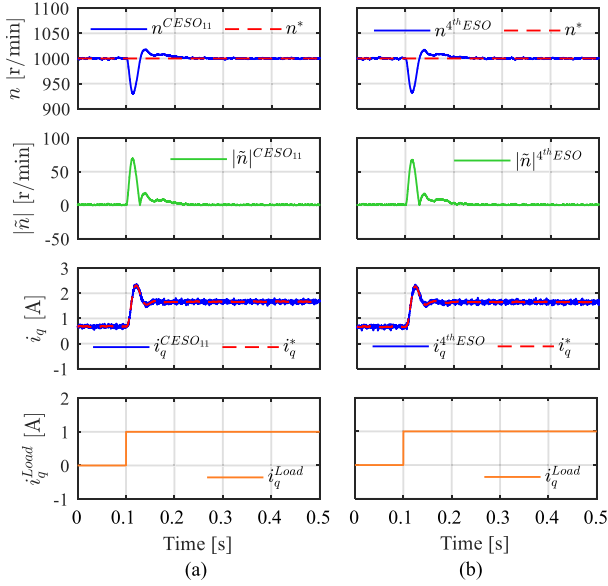


Fig. 16. System response under 1000 r/min with a step load change using different observers. (a) CESO₁₁. (b) Equivalent fourth-order ESO.

As shown in Fig. 16, when subjected to the same external disturbance, both the speed controllers with CESO and its equivalent fourth-order ESO produce identical current references, leading to the same current responses and variations in speed. Hence, their equivalence is demonstrated.

V. CONCLUSION

This article uncovers that the underlying structure of the CESO which comprises two classical second-order ESOs, effectively functions as a fourth-order ESO. Additionally, it reveals that the first observer in the CESO configuration significantly contributes to the disturbance rejection ability. Through a comparative analysis of the properties of different second-order ESOs, two novel CESO configurations with enhanced disturbance rejection ability are proposed. Experimental results demonstrate that the proposed CESO₄₁ exhibits superior disturbance rejection ability compared to the conventional CESO, even when the bandwidth of the conventional configuration is doubled. However, recognizing that disturbance rejection ability is intertwined with noise suppression property, the paper also introduces CESO₁₄ to strike a balance between the two aspects.

APPENDIX

$$\begin{cases}
 \sigma_1 = (1 + G_{11}G_{21} - G_{11} - G_{21})s^4 \\
 \quad + (k_{11} + k_{21} + G_{12}G_{21} - G_{11}k_{21} - G_{21}k_{11})s^3 \\
 \quad + (k_{12} + k_{22} + k_{11}k_{21} - G_{11}k_{22} - G_{21}k_{12})s^2 \\
 \quad + (k_{11}k_{22} + k_{12}k_{21})s \\
 \quad + k_{12}k_{22} \\
 \sigma_2 = (G_{11} + G_{21} - G_{11}G_{21})s^2 \\
 \quad + (G_{11}k_{21} + G_{21}k_{11} - G_{12}G_{21})s \\
 \quad + G_{11}k_{22} + G_{21}k_{12} \\
 \sigma_3 = (G_{12} + G_{22} - G_{11}G_{22})s^2 \\
 \quad + (G_{12}k_{21} + G_{22}k_{11} - G_{12}G_{22})s \\
 \quad + G_{12}k_{22} + G_{22}k_{12} \\
 \sigma_4 = \Delta_{o1}\Delta_{o2}(s + k_p) \\
 \sigma_5 = k_p\sigma_1 + \sigma_3s^2 \\
 \sigma_6 = \Delta_{o1}\Delta_{o2} + (k_p\sigma_2 - \sigma_3)s.
 \end{cases} \quad (A1)$$

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